nature communications

Article

Liquid metal dispersed single-atom catalyst with high-temperature stability

Received: 9 July 2025

Accepted: 27 February 2026

Published online: 13 March 2026

Check for updates

Ziyue Zeng[1,7], Chenyang Wang[1,7], Mingjun Sun[2,7], Dong Liu[3], Donghong Zhang[3], Shiyi He[1], Zijia Liang[4], Yushen Ma[1], Yile Zhang[1], Ling Li[3], Xin Tian[3], Mengqi Zeng[1], Mingyue Ding[3,5], Feng Ding[6] & Lei Fu[1,4]

Single-atom catalysts (SACs) enable greener and more economically sustainable chemical production by significantly improving thermocatalysis efficiency and selectivity through maximized atom utilization and highly homogeneous metal coordination environments. Unfortunately, SACs are fundamentally constrained by the stability owing to the severe aggregation of single atoms, especially under the high-temperature thermocatalysis operations, which compromises the overall catalytic performance. Here, we report a synthetic strategy to realize the highly thermal-stable SACs resistance to sintering at harsh conditions through harnessing the inherent metal affinity and fluidity of liquid metal. A stable liquid metal-active metal interaction is formed, profiting from the superior metal affinity of liquid metal. Combined with the fluidity of liquid metal, active metal atoms can move but remain confined to the liquid metal as the metallic single-atom state at high temperatures. This catalyst exhibits outstanding thermal durability for ethane dehydrogenation, sustaining stable operation for over 100 h at 650 °C with an impressive ethylene selectivity of 98%. The strategy of constructing stable metal-metal interactions by utilizing the inherent metal affinity and dynamic fluidity of liquid metal will pave a practical way for the design of highly thermal-stable SACs.

Industrial thermocatalysis holds great promise for improving the efficiency and sustainability of thermochemical processes and producing value-added chemicals in the modern chemical industry[1–3]. Single-atom catalysts (SACs) have garnered extensive attention for the development of highly efficient, sustainable, cost-effective catalysts in thermocatalysis fields due to their maximized atom utilization and highly homogeneous metal coordination environment[4–8]. However, constructing highly thermally durable SACs is a significant challenge in industrial applications. This is owing to the inherently ultra-high surface energy of single atoms, and the high temperature (>300 °C) provides enough energy to overcome the diffusion barrier of single atoms, leading to the migration of single atoms and severe agglomeration[9]. In the current paradigm, it is prevalent to stabilize SACs by modulating supports (such as carbon-based supports[10], metal oxides characterized by defection[11–14], and so on) that tune interactions between metal atoms and supports. However, the presence of a strong bonding interaction between the supports and the single atoms may result in the compromise of the unique coordination structure of the single atoms, thereby diminishing the catalytic activity and even deactivation[15,16]. In terms of the strategy of using porous supports to spatially confine single atoms (such as molecular sieves and metal-organic frameworks)[17,18], surmounting the substantial surface energy

[1]College of Chemistry and Molecular Sciences, Wuhan University, Wuhan, China. [2]Institute of Technology for Carbon Neutrality, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China. [3]School of Power and Mechanical Engineering, Wuhan University, Wuhan, China. [4]Institute for Advanced Studies, Wuhan University, Wuhan, China. [5]Academy of Advanced Interdisciplinary Studies, Wuhan University, Wuhan, China. [6]Research Division of Advanced Materials, Suzhou Laboratory, Suzhou, China. [7]These authors contributed equally: Ziyue Zeng, Chenyang Wang, Mingjun Sun.
e-mail: dingmy@whu.edu.cn; dingf@szlab.ac.cn; leifu@whu.edu.cn

of the single atoms is still difficult, which will lead to inevitable aggregation of single atoms at elevated temperatures[19,20]. Hence, designing an optimal support that balances metal-support interaction for realizing highly efficient thermal-stable SACs remains an obstacle.

Inspired by the rule of thumb "like dissolves like", it is promising to take advantage of unique metal–metal interaction to efficiently disperse active metals to acquire stable single atoms with inherently high metal activity by using the liquid metal as the dispersion[21–23]. Herein, we achieve the highly efficient dispersion of single atoms with unprecedented thermal stability by utilizing the inherent fluidity and metal affinity of liquid metal. As a demonstration, gallium (Ga), capable of an ideal liquid environment, was employed to spontaneously disperse active metal platinum (Pt). Exceptionally thermal-stable Pt single atoms dispersed by Ga that survived at 650 °C were realized. In this system, profiting from the superior metal affinity of Ga with Pt, Pt–Pt bonds are broken at elevated temperatures, and subsequently the mobile Pt atoms are captured by Ga to form the stable Pt–Ga interaction, accomplishing the highly efficient dispersion of Pt single atoms. The fluidity of liquid metal and the stable Pt–Ga interaction ensure that the Pt atoms can remain dynamically scattered throughout the whole liquid metal dispersion system. Additionally, liquid metal dispersed single atoms can self-update onto the catalysis interface continuously, promoting catalytic reactions at the interface. Hence, these liquid metal dispersed SACs manifested remarkable thermal durability for ethane dehydrogenation (EDH), sustaining stable operation for over 100 h at 650 °C with ethylene selectivity of 98%. Furthermore, the elemental composition of the system was extended to other noble metals, demonstrating the universality of the synthetic strategy. This elucidates the promising potential in realizing highly thermal-stable SACs.

Results

The design of thermal-stable liquid metal dispersed SACs

With increasing temperature, the diffusivity of single atoms is accelerated, thereby enhancing their migration and subsequent aggregation into larger clusters or particles due to thermodynamics favoring lower total surface energy states[9,24]. In essence, the competition of aggregation reflects the competitive bonding between single atom/single atom and single atom/support. Establishing highly efficient as well as stable metal-support interaction is crucial for blocking the aggregation and sintering of single atoms at high temperatures. In this work, we make the realization of thermal-stable SACs by utilizing liquid metal to disperse and stabilize active metals (Fig. 1a). The abundance of metallic bonds in the liquid metal makes it an ideal medium for effectively dispersing active metals with similar metallic bonds, and the active metals can keep the metallic state. At elevated temperatures, enough energy is provided to break metallic bonds in the active metals. Furthermore, the fluidity of liquid metals drives the dispersion of active atoms in liquid metals at elevated temperatures. These dynamically migrating single atoms can be subsequently combined with liquid metal, profiting from its good metallic affinity and forming a stable interaction between these two kinds of atoms. In light of this, equipped with the naturally metallic affinity and fluidity of liquid metal, the highly thermal-stable SACs can be realized.

The liquid metal dispersed SACs are synthesized by dissolving Pt nanoparticles into liquid Ga at elevated temperatures. The mechanism illustrating the formation process of the liquid metal dispersed SACs is shown in Fig. 1a. Firstly, the Pt particles were sprinkled evenly over the liquid metal. As the temperature gradually increases, the solubility of the active metal in the dispersion is further increased, and the active metal can efficiently disperse in the liquid metal within its limited solubility. The Pt–Pt bonds in the original Pt metal nanoparticles are broken and tend to bind tightly to Ga atoms and form a thermodynamically stable configuration with Ga atoms, benefiting from the good affinity between Ga and Pt. The stable Pt–Ga interaction curtails the sintering of Pt single atoms. Moreover, according to the Arrhenius equation[25], the thermal motion of atoms would be intensified by high temperature, promoting the rapid diffusion between Pt and Ga atoms and the continuous renewal of Pt single atoms exposed on the catalytic surface (Fig. 1b). In contrast, when single atoms are anchored to solid supports, their co-migration with the supports becomes kinetically constrained. Once the metal-support interactions become insufficient to compensate for the inherently high surface energy of single atoms, these atoms tend to detach from the solid supports and aggregate. This process is exothermic, where minimization of total surface energy governs particle growth[26]. Meanwhile, some single atoms may embed into the lattice of supports, becoming buried when the interaction between the support and metal atoms is too strong. These lead to the loss of active sites, reducing the stability of SACs and consequently affecting their activity and selectivity (Fig. 1c).

Thermal stability of active metal atoms mono-dispersed in liquid Ga

Ab initio molecular dynamics (AIMD) simulations were employed to elucidate the atomic-scale dispersed state of active metal in liquid metal (computational details are provided in the Supplementary information). Using Pt_{10} as a model of Pt cluster, we investigated the structural evolution of both clustered and dispersed Pt_{10} in liquid Ga during 10 ps simulations at 923 K (Fig. 2a, b, and Supplementary Movies 1, 2). The calculated total energy versus time curves for the $Ga_{90}Pt_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10} (Fig. 2c) reveal that the dispersed Pt_{10} generally exhibits lower total energy compared to the aggregated Pt_{10} clusters within the simulated 10 ps. The results indicate that the remarkable atomic diffusion of liquid Ga metal at high temperature promotes the migration of heterogeneous active metal atoms, thereby stabilizing their dispersed atomic state in liquid Ga.

The relatively high Pt concentration in the $Ga_{90}Pt_{10}$ system leads to intermittent formation of Pt_2 dimer, as evidenced by the time-dependent curve of the shortest Pt–Pt distance (Supplementary Fig. 1). To further elucidate the actual existence state of Pt atoms in liquid Ga, we performed AIMD simulations of the $Ga_{98}Pt_2$ system (a Pt_2 dimer embedded in Ga_{98}) at 923 K. The 10 ps NVT ensemble simulations (Fig. 2d) display that the Pt–Pt interatomic distance progressively increases from 4.18 to 11.63 Å, significantly exceeding the Pt–Pt bond length of 2.65–2.79 Å[27]. This unequivocally confirms that Pt atoms preferentially exist as monatomic species in the liquid Ga (Supplementary Fig. 2 and Supplementary Movies 3, 4). Further statistical analysis of the coordination environment of Pt atoms in the $Ga_{98}Pt_2$ system shows that the dominant coordination numbers for individual Pt atoms are 8 or 9 (Supplementary Figs. 3 and 4), implying that mono-dispersed Pt atoms migrate within liquid Ga in the form of clusters, such as Ga_8Pt or Ga_9Pt (Supplementary Fig. 5).

To fundamentally understand the thermodynamic preference for Pt mono-dispersion in liquid Ga, we systematically calculated the bond energy difference ΔE_b ($\Delta E_b = E_{(Ga_nPt)_2(Pt-Pt)} + \bar{E}_{Ga_n(Ga-Ga)} - 2E_{Ga_{n+1}Pt(Ga-Pt)}$) only for the Pt_2, $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters (Fig. 2e and Supplementary Figs. 6–11). Here, $\bar{E}_{Ga_n(Ga-Ga)}$ represents the average Ga–Ga bond energy in the Ga_n molecular clusters, $E_{(Ga_nPt)_2(Pt-Pt)}$ denotes the Pt–Pt bond energy in the $(Ga_nPt)_2$ molecular clusters, and $E_{Ga_{n+1}Pt(Ga-Pt)}$ refers to the Ga–Pt bond energy in the $Ga_{n+1}Pt$ molecular clusters. The results indicate that the ΔE_b values for the Pt_2, $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters, evaluated using different computational methods (B3LYP, MN15L, and M06L), are all greater than zero. This demonstrates that in these molecular systems, the Ga atoms involved in bond-breaking readily disrupt the Pt–Pt bonds, leading to the formation of stable Pt–Ga–Pt bridging units, thereby promoting the single-atom dispersion of Pt in liquid Ga. Meanwhile, we calculated the formation energies of Pt_2Ga, $(Ga_6Pt)_2Ga$, and $(Ga_7Pt)_2Ga$ molecules,

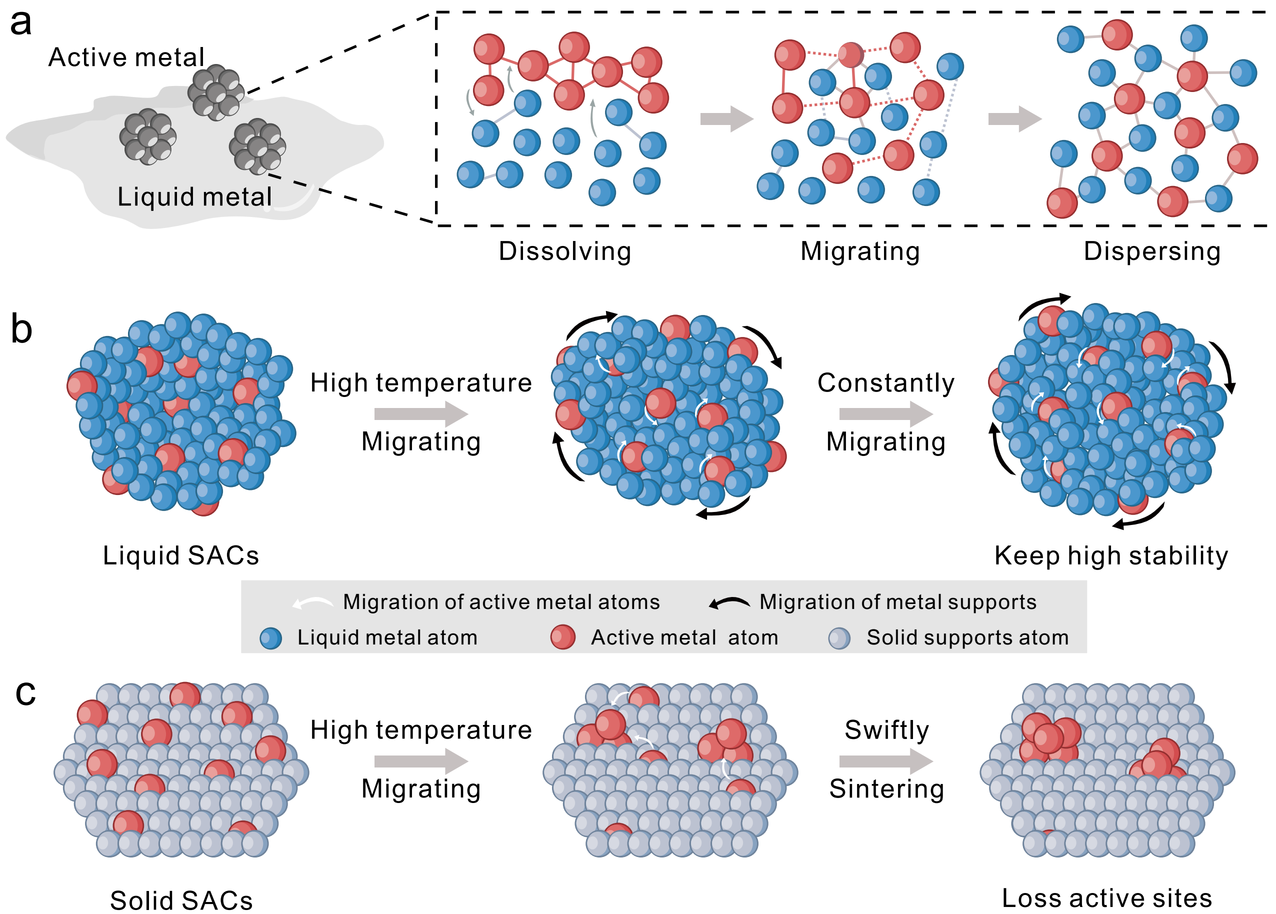

Fig. 1 | The design of thermal-stable liquid metal dispersed SACs. a The schematic diagram for the synthesis of liquid metal dispersed SACs. The migration of active single atoms in liquid metal (**b**) and on traditional solid support (**c**).

which were determined to be 101.1, 46.2, and 57.9 kcal mol^{-1}, respectively (Supplementary Fig. 12 and Fig. 2f). Additionally, we roughly estimated the energy barriers of Ga atom insertion into the Pt–Pt bond to be 58.2, 13.4, and 40.2 kcal mol^{-1} by performing rigid scans of the Pt–Ga–Pt angle (Supplementary Fig. 13 and Fig. 2f). These results clearly demonstrate that, both thermodynamically and kinetically, Ga atoms in liquid Ga at 923 K can readily disrupt Pt–Pt bonds, thereby promoting the mono-dispersion of Pt atoms.

Characterization of liquid catalyst

To further validate that liquid metal can act as the dispersion for single atoms, we synthesized the Ga-dispersed Pt SACs (Pt SACs@Ga). The detailed structure of the catalyst was characterized by transmission electron microscopy (TEM) and energy dispersive spectrometry (EDS). The Pt species in the liquid metal exist in a mono-disperse distribution without the obvious occurrence of aggregated Pt (Fig. 3a). The bright-field image of Pt SACs@Ga liquid catalyst maintains an almost invariant brightness across the whole particle and the corresponding selected area electron diffraction (SAED) data shows no diffraction spots of Pt in the amorphous ring, revealing the lack of long-range order structure in the Ga–Pt catalyst (Supplementary Fig. 21 and Fig. 3b). Furthermore, the X-ray photoelectron spectroscopy (XPS) spectrum was employed to reveal the chemical composition and electronic structure of the liquid metal dispersed SACs. Before the execution of XPS measurements, the sample underwent etching with argon ions to eliminate the

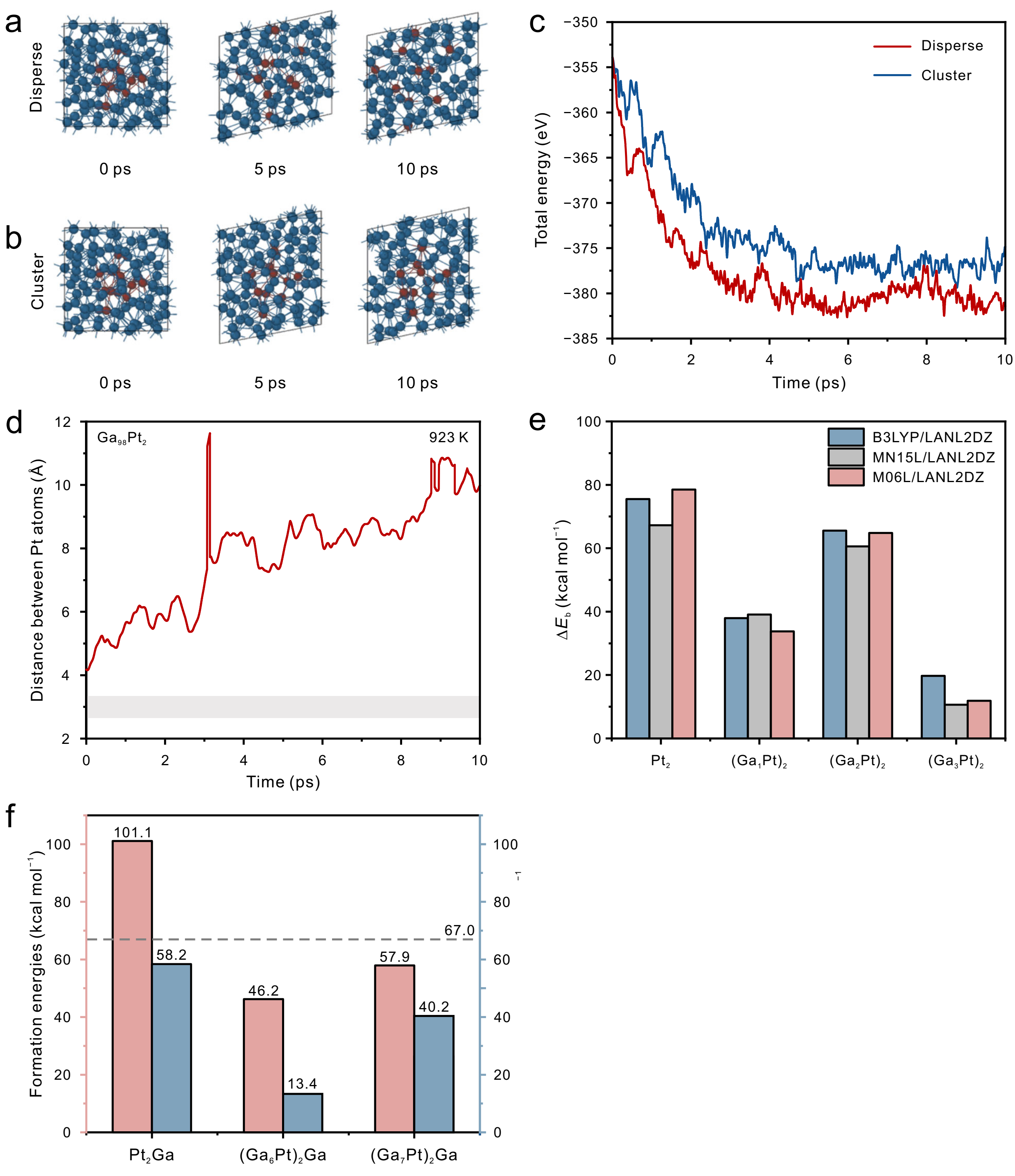

Fig. 2 | Theoretical calculations of heterogeneous active metal atoms monodispersed in liquid Ga. The extracted 0, 5, and 10 ps snapshots from AIMD trajectories of both dispersed (**a**) and clustered (**b**) Pt_{10} in liquid Ga. **c** Total energy versus time curves for the $Ga_{90}Pt_{10}$ system containing clustered Pt_{10} and dispersed Pt_{10}. **d** Time evolution of Pt–Pt interatomic distance in the $Ga_{98}Pt_2$ system under NVT ensemble simulation. **e** The calculated bond energy difference ΔE_b using different computational methods (B3LYP, MN15L, and M06L) for the Pt_2, $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters. **f** The formation energies and energy barriers associated with Ga atom insertion into Pt–Pt bonds in Pt_2Ga, $(Ga_6Pt)_2Ga$, and $(Ga_7Pt)_2Ga$ molecular systems (the rough estimation suggests that the reaction can overcome an energy barrier of 67 kcal mol^{-1} at 923 K[42])

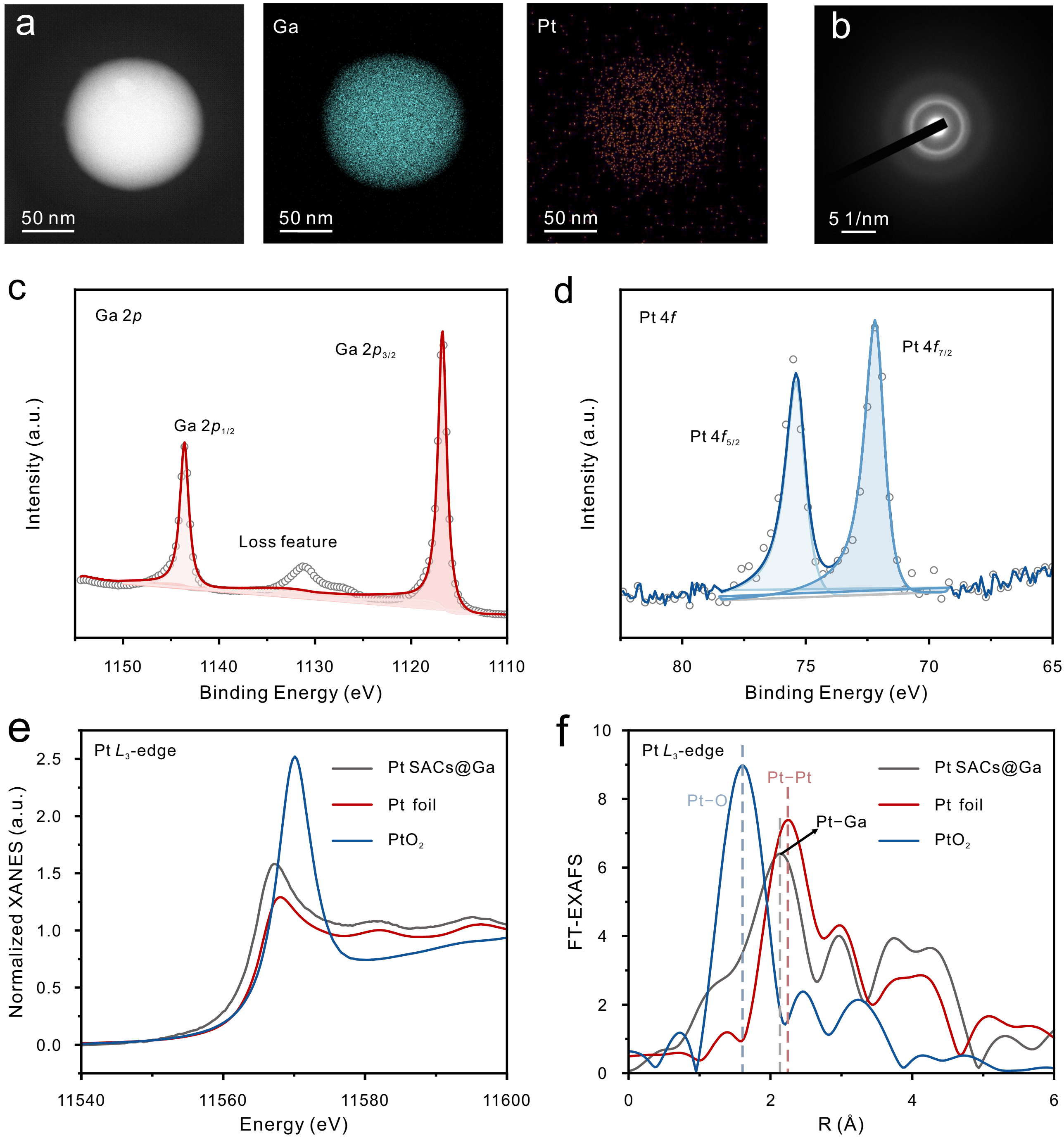

Fig. 3 | Characterization of liquid catalyst. a TEM image and EDS elements mapping of the liquid catalyst. **b** SAED pattern of the liquid catalyst, XPS spectra of Ga 2*p* (**c**) and Pt 4*f* (**d**) in the liquid catalyst. **e** Normalized XANES spectra at the Pt L_3-edge. **f** FT Pt L_3-edge EXAFS spectra in *R* space of liquid catalyst with phase correction.

native oxide layer and surface contaminants. Ga 2*p* peaks appear at 1116.73 and 1116.59 eV (Fig. 3c), which are consistent with the reported binding energies of metallic Ga (Ga^0 $2p_{3/2}$ = 1116.70 eV). Meanwhile, due to the higher electronegativity of Pt (2.28) than that of Ga (1.81)[28], the Pt in the Pt SACs@Ga system serves as the electron acceptor, and the Pt $4f_{7/2}$ and $4f_{5/2}$ peaks in Pt SACs@Ga exhibit a positive shift (1.0 eV) than those in pure Pt metal (Fig. 3d). It is a consensus that metallic Pt^0 species occupy a preeminent role as the active sites for many catalytic processes[29–31]. This guarantees a proficient and robust distribution of single atoms while preserving their catalytic activity. Furthermore, the absence of the obvious aggregation of Pt atoms in the Pt SACs@Ga is further identified by X-ray diffraction (XRD) tests, where no signal of Pt (PDF#04-0802) is observed (Supplementary Fig. 22). To mitigate the potential inaccuracies arising from the insensitivity of XRD to small Pt clusters, the pair distribution function (PDF) analysis was conducted to delve into the interatomic structure of the Pt SACs@Ga liquid catalyst (Supplementary Fig. 23). The absence of Pt−Pt bonds indicates that Pt is distributed nearly at the atomic level within the liquid Ga without the formation of Pt clusters. These observations suggest that liquid metal can achieve the efficient uniform dispersion of highly active single atoms at elevated temperatures.

To further verify the exact dispersion state of Pt in the Pt SACs@Ga catalyst, X-ray absorption spectroscopy (XAS) measurements were conducted. The X-ray absorption near-edge structure

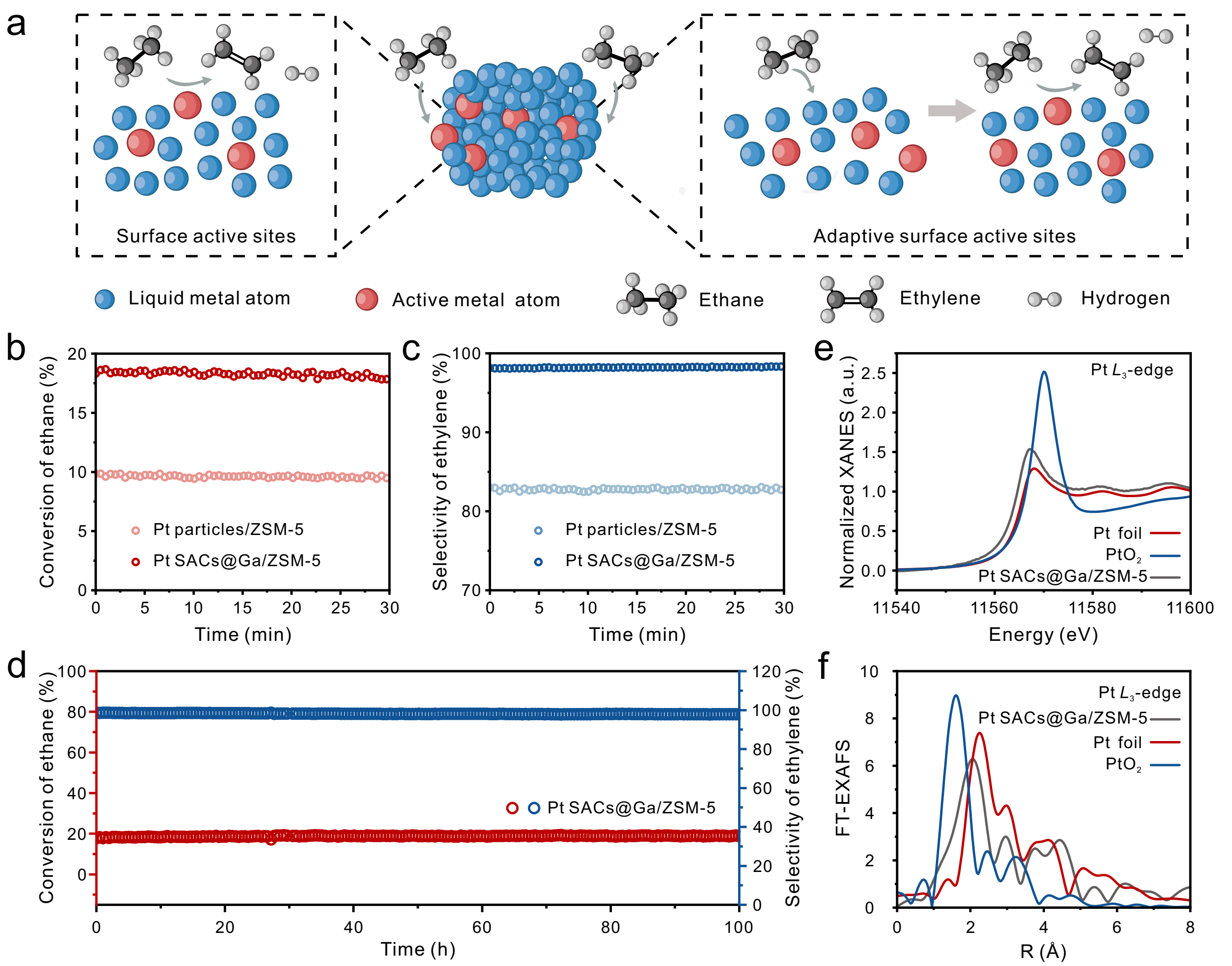

Fig. 4 | EDH performance of liquid metal-dispersed SACs. a Schematic showing the structure evolution of adaptive surface active sites in liquid metal dispersed SACs when exposed to ethane. **b, c** Catalytic performances at 650 °C. **d** Long-term stability test. **e** XANES spectra of the liquid metal dispersed SACs after 100 h EDH taken at Pt L_3 edge. **f** FT spectra in *R*-space at Pt L_3 edge.

(XANES) spectrum shows that the intensity of the white line at the L_3 edge of Pt lies between that of the Pt foil and that of PtO_2. Correspondingly, the k_3-weighted *R*-space Fourier transformed spectra of the extended X-ray absorption fine structure (EXAFS) show the prominent peak at 2.14 Å for Pt SACs@Ga distinguished from that of the Pt–O (1.62 Å) bonds for PtO_2 and the Pt–Pt bonds (2.24 Å) for Pt foil (Fig. 3f). All these results intuitively indicate that the Pt exists in the single-atom state in liquid metal.

EDH performance of liquid metal-dispersed SACs

Based on the computational results, the liquid metal-dispersed SACs theoretically feature excellent thermal stability at high temperatures. It is necessary to conduct specific high-temperature tests for verification. Pt-based catalysts are used for the dehydrogenation of alkanes due to their high efficiency in activating the C–H bonds of alkanes and low propensity in breaking C–C bonds[32,33]. And Pt–Sn/Al_2O_3 (Oleflex process) is the primary industrial technology operated for direct propane dehydrogenation[34]. While it suffers from severe sintering and coke, which results in rapid deactivation and restricts the selectivity[35]. Therefore, as a proof of concept, the thermal stability of liquid metal-dispersed SACs was validated by EDH. As shown in Fig. 4a, when exposed to ethane, Pt single atoms exposed on the surface capture ethane molecules and synergistically activate C–H bonds, leading to selective EDH to ethylene. Meanwhile, the fluidic characteristic of liquid metal dispersed SACs at operating conditions enables continuous replenishment of catalytically active Pt atoms onto the surface. The stable interaction between Pt–Ga effectively blocks the aggregation of Pt atoms during the constant migration. Benefiting from the advantages mentioned above, these adaptive surface active sites maintain long-term activity and thermal stability.

To circumvent the issue of inadequate interaction between the liquid metal dispersed SACs and the reaction gas, the porous support is utilized to amplify the gas mass transfer rate. So, we coupled liquid metal dispersed SACs with porous MFI zeolite (ZSM-5) by the physical approach of ultrasonication to increase the contact surface area between the liquid metal support SACs and gas reactants. As demonstrated in Supplementary Fig. 24, EDS mappings revealed a well-distribution of Ga, Si, and Al, indicating the liquid metal is uniformly loaded in the ZSM-5 zeolite framework. Upon heating, Pt will be spontaneously dispersed in Ga homogeneously, achieving the preparation of a zeolite-dispersed liquid metal supported SACs (Pt SACs@Ga/ZSM-5). Further, we performed in situ XRD to explore the dispersing process of Pt (Supplementary Fig. 25). The diffraction peak observed at 43–47° originates from the carbon in situ reactor (Supplementary Fig. 26). Initially, the characteristic peaks of Pt can be apparently observed, which are located at 39.76° and 46.24°. With the temperature increasing, the characteristic peaks of Pt were gradually weakened and disappeared subsequently, proving the dispersion of Pt particles.

The catalytic performance was evaluated in non-oxidative EDH at 923 K in a fixed-bed quartz U-tube reactor under atmospheric pressure with a feed gas of 12.5 vol% C_2H_6 in Ar. The Pt particles/ZSM-5 catalyst, which is synthesized by direct mixing commercial Pt particles with ZSM-5, exhibits ethane conversion at 9.5% and ethylene selectivity at 82.6% (Fig. 4b, c). Interestingly, the liquid metal dispersed SACs are highly active with ethane conversion at 17.8% and ethylene selectivity at 98.3%. This is because the introduction of liquid metal alters the state of Pt. By dispersing Pt particles into single Pt atoms, the utilization of Pt atoms is maximized to achieve high conversion rates. Furthermore, since the size of Pt particle significantly influences catalytic performance, larger Pt nanoparticles typically promote C–C cleavage and deep dehydrogenation side reactions[36]. In contrast, Pt-based catalysts dispersed at the atomic scale exhibit excellent selectivity for ethylene due to their superior activation of C–H bonds in alkanes, while simultaneously limiting C–C bond cleavage to prevent dehydrogenation reactions. Therefore, when liquid metal is introduced, Pt exists in a single-atom state, enabling the Pt SACs@Ga/ZSM-5 catalyst to achieve high selectivity. To verify the universality of the strategy that liquid metals can disperse metals into single atoms, we extended the synthesis approach to other systems and successfully synthesized liquid metal-dispersed Rh SACs (Supplementary Fig. 27). Moreover, Rh SACs@Ga/ZSM-5 exhibits significantly improved catalytic performance compared to Rh particles/ZSM-5 in EDH (Supplementary Fig. 28).

To gain a deeper understanding of the effects of liquid metal-dispersed SACs and nanoparticles catalysts on the dehydrogenation reaction of ethane, we conducted kinetic studies on both types of catalysts. We measured the ethane conversion in a temperature range of 883–933 K and fitted the data according to the Arrhenius equation to calculate the apparent activation energy for EDH. The apparent activation energies for Pt SACs@Ga/ZSM-5 and Pt particles/ZSM-5 are 102.44 and 137.18 kJ mol^{-1}, respectively (Supplementary Fig. 29). The significantly reduced activation energy indicates a decreased energy barrier that needs to be overcome for the reaction to proceed. And the single Pt atom dispersed in the liquid Ga can more effectively weaken the C–H bond of ethane than Pt particles. Therefore, the Pt SACs@Ga/ZSM-5 catalyst exhibits superior catalytic performance.

The performance of the liquid metal-dispersed SACs in the durability test is presented in Fig. 4d. It exhibits sustainable conversion and selectivity in the continuous reaction for 100 h, where the constant ethane conversion of ~18% is obtained without attenuation. And liquid metal dispersed SACs exhibit a steady ethylene productivity of around 14.3 g g_{Pt}^{-1} h^{-1} during the 100 h EDH test. The catalyst displays no deactivation trend during the testing period, indicating a reasonable expectation of more durable stability. Notably, the XANES spectrum and *R*-space of liquid metal dispersed SACs after 100 h EDH at Pt L_3-edge are similar to those of pristine liquid SACs (Figs. 4e, f, and 3e, f), demonstrating exceptional structural integrity under prolonged thermal catalysis. The catalytic performance and structural characterization together confirm the superior thermal stability of liquid metal dispersed SACs at extremely high temperatures, which remarkably outperforms the traditional solid support loaded SACs. To further substantiate the thermal stability performance of our work, comparing the catalytic reaction temperatures and time on stream of Pt SACs@Ga/ZSM-5 with reported SACs (Supplementary Table 1), where the results remain outstanding. The excellent stability, selectivity, and conversion indicate the potential of liquid metal-dispersed SACs for reactions limited in harsh conditions.

Discussion

We succeeded in achieving the highly thermal-stable SACs by harnessing the inherent fluidity and metal affinity of liquid metal to form stable metal–metal interaction at high temperatures. In this system, the Pt–Pt bond in the active metal is broken at elevated temperatures, and then Pt atoms are captured by liquid metal to form a highly stable Pt–Ga interaction, which enables Pt atoms to constantly remain uniformly dynamically decentralized to liquid metal. Crucially, the Pt atoms can maintain a highly stable dispersed state during EDH at 650 °C for 100 h, while the catalyst maintains an impressive selectivity of 98%. Furthermore, this approach can be applied to other material systems to acquire SACs. Our strategy provides accessible guidance for acquiring highly thermal-stable SACs by using liquid metal as the dispersion.

Methods

Chemicals and materials

Gallium (Ga, 99.999%) was purchased from Shanghai Minor Metals Co. Ltd., platinum (Pt, 99.9%) and molecular sieve (ZSM-5 zeolites, Si/Al = 45) were purchased from Aladdin Industrial Inc. Rhodium (Rh, 99.95%) powder was purchased from Shanghai Macklin Biochemical Co., Ltd. High-purity ethane (99.99%) was purchased from Wuhan Xiangyun Gas Technology Co., Ltd. and was purified using 4A zeolite before reaction.

Catalyst synthesis

Synthesis of liquid metal dispersed SACs. The liquid metal dispersed SACs were synthesized by dissolving Pt or Rh nanoparticles into molten Ga in a tube furnace in a H_2/Ar mixture gas at 650 °C, with a heating rate of 5 °C min^{-1}.

Synthesis of the catalyst for EDH. In order to increase the mass transfer of the ethane, the liquid metal was coupled with the ZSM-5 (Ga/ZSM-5) by sonication as follows. The liquid Ga and ZSM-5 zeolites were transferred to ethylene glycol, and the solution was sonicated by a probe-type sonicator for 2 h (5 s on and 3 s off) at a power of 290 W. After sonication, the resultant precipitations were washed with ethanol and water 3 times. And then the resultant was flash frozen in liquid nitrogen for 5 min and freeze-dried at −40 °C, under a pressure less than 200 mTorr, for 24 h using the FreeZone 4.5 liter benchtop freeze dry system. The Pt particles were mixed with Ga/ZSM-5 (or pure ZSM-5) and heated from room temperature to 650 °C at a rate of 10 °C min^{-1} in the reactor under a H_2/Ar mixed gas atmosphere. The mixture was then held at 650 °C for 30 min to complete the synthesis of the Pt SACs@Ga/ZSM-5 catalyst (or Pt particles/ZSM-5). The Pt : Ga atomic ratio of the Pt SACs@Ga/ZSM-5 catalyst prepared for catalytic testing was maintained at 0.2 : 99.8. After purging the H_2 from the U-tube, the catalyst was ready for catalytic testing. Similarly, replace the Pt particles with Rh particles and undergo the same treatment to obtain Rh SACs@Ga/ZSM-5 (or Rh particles/ZSM-5), and the Rh : Ga atomic ratio of the Rh SACs@Ga/ZSM-5 catalyst used for catalytic testing was also maintained at 0.2 : 99.8.

Catalyst characterizations

The morphology of the liquid metal dispersed SACs was characterized by an F200 transmission electron microscope (TEM) (JEOL, Japan) operated at 200 kV. X-ray diffraction (XRD) data was employed by PANalytical X'Pert-Pro diffractometer, with a Cu Kα radiation (λ = 0.15406 nm), in the 2θ scan mode ranging from 10° to 80° at a scan speed of 8° min^{-1}. In situ XRD was collected with the Rigaku Ultima IV instrument using Cu Kα (λ = 1.5406 Å) radiation at 40 kV and 40 mA, respectively. The catalysts were scanned at a rate of 8° min^{-1} in the 2θ range of 35°–60°. Pair distribution function (PDF) analysis was performed at room temperature in an air atmosphere by SmartLab, with a Ag Kα radiation (λ = 0.056088 nm). X-ray photoelectron spectroscopy (XPS) measurements were taken on a high-resolution Thermo Fisher Scientific ESCALAB Xi+ spectrometer using Al Kα radiation under vacuum. Pt L_3 and Rh *K*-edge XAFS spectra were collected at the BL11B beamline of the Shanghai Synchrotron Radiation Facility (SSRF).

During the measurement, the synchrotron radiation ring was operated at 3.5 GeV, and the electron current was 200 mA in the top-up mode. The samples were sealed in Kapton film. For the measurement of Pt L_3 edge, the white light was monochromatized by a Si (111) double-crystal monochromator. And the white light was monochromatized by a Si (311) double-crystal monochromator when the Rh K edge was measured. The corresponding metal foils were used as the reference samples to calibrate and measure in the transmission mode. All XAFS spectra were processed using the IFEFFIT package. The elemental composition of the samples before and after the reaction was determined by inductively coupled plasma-atomic emission spectroscopy (ICP-OES).

Catalytic performance and kinetic behavior measurements

EDH tests were performed under atmospheric pressure in a fixed-bed quartz U-tube reactor. Catalysts were weighed and loaded into 10 mm outside diameter, 8 mm interior diameter U-tube reactor. 5 sccm of argon and 15 sccm of hydrogen were flowed at 650 °C for 10 min to reduce any metal oxides. And then the reactor was purged with 20 sccm argon for 5 min to prevent hydrogen residue. The EDH reaction was conducted in a C_2H_6/Ar mixture gas (Vol C_2H_6/Vol Ar = 1:7) with a flow rate of 20 mL min^{-1}. The outlet gas products were observed by an online mass spectrometer (Pro-tech, PM-QMS, China). The conversion of ethane and selectivity to all carbonaceous products (including coke) were calculated as follows:

$$Ethane\ conversion = \frac{\mathrm{molar}\ flow\ rate\ of\ carbon\ (C)\ in\ products}{\mathrm{molar}\ flow\ rate\ of\ total\ \mathrm{C}\ in\ effluent\ (C_2\mathrm{H}_6 + products)} \times 100\% \tag{1}$$

$$Product\ selectivity = \frac{\mathrm{molar}\ flow\ rate\ of\ C\ in\ product\ i}{\mathrm{molar}\ flow\ rate\ of\ C\ in\ all\ products} \times 100\% \tag{2}$$

The kinetic behavior of the catalyst for EDH needs to minimize transport limitations. Therefore, the conversion was limited to 5–15% while ethylene selectivity remained above 90% by systematically varying the catalyst amount. By analyzing the temperature and pressure dependence of these rates, we extracted the activation energy and reaction order for the reaction.

method[40] was used to evaluate the Pt–Pt bonding in liquid Ga. To gain further insight into the dispersion of Pt atoms in liquid Ga, we employed the B3LYP/LANL2DZ, MN15L/LANL2DZ[41], and M06L/LANL2DZ methods to calculate the bond energy difference ΔE_b. The specific calculation method of ΔE_b for the Pt_2, $(Ga_1Pt)_2$, $(Ga_2Pt)_2$, and $(Ga_3Pt)_2$ molecular clusters is based on Eq. 3:

$$\Delta E_b = E_{(\mathrm{Ga_nPt})_2(\mathrm{Pt-Pt})} + \bar{E}_{\mathrm{Ga_n(Ga-Ga)}} - 2E_{\mathrm{Ga_{n+1}Pt(Ga-Pt)}} \tag{3}$$

Here, $\bar{E}_{\mathrm{Ga}_n\mathrm{(Ga-Ga)}}$ represents the average Ga–Ga bond energy in the Ga_n molecular clusters, $E_{(\mathrm{Ga_nPt})_2(\mathrm{Pt-Pt})}$ denotes the Pt–Pt bond energy in the $(Ga_nPt)_2$ molecular clusters, and $E_{\mathrm{Ga_{n+1}Pt(Ga-Pt)}}$ refers to the Ga–Pt bond energy in the $Ga_{n+1}Pt$ molecular clusters. All bond energy calculations were corrected for basis set superposition error (BSSE). Furthermore, we constructed cluster structures, such as Pt_2, $(Ga_6Pt)_2$, $(Ga_7Pt)_2$, Pt_2Ga, $(Ga_6Pt)_2Ga$, and $(Ga_7Pt)_2Ga$, optimizing all cluster molecules using the B3LYP/LANL2DZ method. Additionally, we performed rigid scans of the Pt–Ga–Pt angle in the Pt_2Ga, $(Ga_6Pt)_2Ga$, and $(Ga_7Pt)_2Ga$ cluster systems to roughly estimate the energy barrier associated with the incorporation of Ga atoms into the Pt–Pt bond.

Data availability

All data are available from the corresponding author upon request. The relevant data underlying the figures of this study are available with the paper. Source data are provided in this paper. Source data are provided with this paper.

References

1. Peng, M. et al. Thermal catalytic reforming for hydrogen production with zero CO_2 emission. *Science* **387**, 769–775 (2025).
2. Wang, Y., Zhang, Y., Wang, X., Liu, Y. & Wu, Z. Photothermal direct methane conversion to formaldehyde at the gas-solid interface under ambient pressure. *Nat. Commun.* **16**, 2550 (2025).
3. Almallahi, R., Wortman, J. & Linic, S. Overcoming limitations in propanedehydrogenation by codesigning catalyst-membrane systems. *Science* **383**, 1325–1331 (2024).
4. Li, M. et al. Structural design of single-atom catalysts for enhancing petrochemical catalytic reaction process. *Adv. Mater.* **36**, 2313661 (2024).
5. Liang, X., Fu, N., Yao, S., Li, Z. & Li, Y. The progress and outlook of metal single-atom-site catalysis. *J. Am. Chem. Soc.* **144**, 18155–18174 (2022).
6. Kment, Š. et al. Single atom catalysts based on earth-abundant metals for energy-related applications. *Chem. Rev.* **124**, 11767–11847 (2024).
7. Peng, M. et al. Interfacial catalysis at atomic level. *Chem. Rev.* **125**, 2371–2439 (2025).
8. Zeng, L. et al. Stable anchoring of single rhodium atoms by indium in zeolite alkane dehydrogenation catalysts. *Science* **383**, 998–1004 (2024).
9. Wei, S. et al. Direct observation of noble metal nanoparticles transforming to thermally stable single atoms. *Nat. Nanotechnol.* **13**, 856–861 (2018).
10. Li, Y. et al. Multiscale anion-hybrid in atomic ni sites for high-rate water electrolysis: insights into the charge accumulation mechanism. *J. Am. Chem. Soc.* **146**, 14194–14202 (2024).
11. Lang, R. et al. Single-atom catalysts based on the metal-oxide interaction. *Chem. Rev.* **120**, 11986–12043 (2020).
12. Shen, Y. et al. Room-temperature photosynthesis of propane from CO_2 with Cu single atoms on vacancy-rich TiO_2. *Nat. Commun.* **14**, 1117 (2023).
13. Zhang, J., Yang, X., Xu, G., Biswal, B. K. & Balasubramanian, R. Accumulation of long-lived photogenerated holes at indium single-atom catalysts via two coordinate nitrogen vacancy defect engineering for enhanced photocatalytic oxidation. *Adv. Mater.* **36**, 2309205 (2024).
14. Wei, S. et al. Self-carbon-thermal-reduction strategy for boosting the fenton-like activity of single Fe-N_4 sites by carbon-defect engineering. *Nat. Commun.* **14**, 7549 (2023).
15. Li, X. et al. Functional CeO_x nanoglues for robust atomically dispersed catalysts. *Nature* **611**, 284–288 (2022).
16. Pu, T., Zhang, W. & Zhu, M. Engineering heterogeneous catalysis with strong metal-support interactions: characterization, theory and manipulation. *Angew. Chem. Int. Ed.* **62**, e202212278 (2022).
17. Liu, H. et al. Encapsulation of Pd single-atom sites in zeolite for highly efficient semihydrogenation of alkynes. *J. Am. Chem. Soc.* **146**, 24033–24041 (2024).
18. Mo, Q. et al. Engineering single-atom sites into pore-confined nanospaces of porphyrinic metal-organic frameworks for the highly efficient photocatalytic hydrogen evolution reaction. *J. Am. Chem. Soc.* **144**, 22747–22758 (2022).
19. Wang, T. et al. Nature of metal-support interaction for metal catalysts on oxide supports. *Science* **386**, 915–920 (2024).
20. Lang, Z. et al. Destabilization of single-atom catalysts: characterization, mechanisms, and regeneration strategies. *Adv. Mater.* **37**, 2418942 (2025).
21. Taccardi, N. et al. Gallium-rich Pd–Ga phases as supported liquid metal catalysts. *Nat. Chem.* **9**, 862–867 (2017).
22. Rahim, M. A. et al. Low-temperature liquid platinum catalyst. *Nat. Chem.* **14**, 935–941 (2022).
23. Tang, J. et al. Dynamic configurations of metallic atoms in the liquid state for selective propylene synthesis. *Nat. Nanotechnol.* **19**, 306–310 (2023).
24. Campbell, C. T., Parker, S. C. & Starr, D. E. The effect of size-dependent nanoparticle energetics on catalyst sintering. *Science* **298**, 811–814 (2002).
25. Laidler, K. J. The development of the Arrhenius equation. *J. Chem. Educ.* **61**, 494–498 (1984).
26. Campbell, C. T. The energetics of supported metal nanoparticles: relationships to sintering rates and catalytic activity. *Acc. Chem. Res.* **46**, 1712–1719 (2013).
27. Dedieu, A. & Hoffmann, R. Platinum(0)-platinum(0) dimers. Bonding relationships in a d^{10}-d^{10} system. *J. Am. Chem. Soc.* **100**, 2074–2079 (1978).
28. Hao, J. et al. Unraveling the electronegativity-dominated intermediate adsorption on high-entropy alloy electrocatalysts. *Nat. Commun.* **13**, 2662 (2022).
29. Cheng, N. et al. Platinum single-atom and cluster catalysis of the hydrogen evolution reaction. *Nat. Commun.* **7**, 13638 (2016).
30. Yu, F.-Y. et al. Pt–O bond as an active site superior to Pt^0 in hydrogen evolution reaction. *Nat. Commun.* **11**, 490 (2020).
31. Dun, Y. et al. Pt^0-$MnSO_4$ active centers on modified $SmMn_2O_5$ mullite oxides for efficient propane oxidation. *Appl. Catal. B* **371**, 125223 (2025).
32. Sattler, J. J. H. B., Ruiz-Martinez, J., Santillan-Jimenez, E. & Weckhuysen, B. M. Catalytic dehydrogenation of light alkanes on metals and metal oxides. *Chem. Rev.* **114**, 10613–10653 (2014).
33. Newmeyer, E.-R. et al. Plasmonic charge localization and C–H activation at single-atom sites in dilute copper platinum alloys. *J. Am. Chem. Soc.* **147**, 11789–11799 (2025).
34. Searles, K. et al. Highly productive propane dehydrogenation catalyst using silica-supported Ga–Pt nanoparticles generated from single-sites. *J. Am. Chem. Soc.* **140**, 11674–11679 (2018).
35. Cheng, Q. et al. Highly efficient catalytic propane dehydrogenation driven by MFI zeolite defect sites. *Nat. Commun.* **16**, 5696 (2025).
36. Zhang, W. et al. Size dependence of Pt catalysts for propane dehydrogenation: from atomically dispersed to nanoparticles. *ACS Catal.* **10**, 12932–12942 (2020).
37. Perdew, J. P. et al. Restoring the density-gradient expansion for exchange in solids and surfaces. *Phys. Rev. Lett.* **100**, 136406 (2008).
38. Parrinello, M. & Rahman, A. Polymorphic transitions in single crystals: a new molecular dynamics method. *J. Appl. Phys.* **52**, 7182–7190 (1981).
39. Parrinello, M. & Rahman, A. Crystal structure and pair potentials: a molecular-dynamics study. *Phys. Rev. Lett.* **45**, 1196–1199 (1980).
40. Pan, H. et al. Benchmarking coordination number prediction algorithms on inorganic crystal structures. *Inorg. Chem.* **60**, 1590–1603 (2021).
41. Yu, H. S., He, X. & Truhlar, D. G. MN15-L: a new local exchange-correlation functional for Kohn-Sham density functional theory with broad accuracy for atoms, molecules, and solids. *J. Chem. Theory Comput.* **12**, 1280–1293 (2016).
42. Shaydullin, R. R., Galushko, A. S., Ilyushenkova, V. V., Vlasova, Y. S. & Ananikov, V. P. Are activation barriers of 50–70 kcal mol^{-1} accessible for transformations in organic synthesis in solution? *Chem. Sci.* **16**, 5289–5298 (2025).

Acknowledgements

The research was supported by the Natural Science Foundation of China (22025303 to L.F., 22502148 to C.Y.W., and 52572035 to M.Q.Z.), the

Fundamental Research Funds for the Central Universities (2042025kf0007 to L.F.), the Postdoctoral Fellowship Program of CPFS (GZB20240567 to C.Y.W.), the cooperation and exchange program of the National Natural Science Foundation of China (22461160283 to F.D.), and the research program from Suzhou Laboratory (SK-1502-2024-055 to F.D.). We thank the Center for Electron Microscopy at Wuhan University for their substantial support of TEM work. We thank the Core Facility of Wuhan University for the measurement of inductively coupled plasma-atomic emission spectrometry, XPS, TEM, and the Core Research Facilities of the College of Chemistry and Molecular Sciences at Wuhan University for the XRD and PDF characterizations. We also thank the BL11B beamline of the Shanghai Synchrotron Radiation Facility for the XAFS characterization.

Author contributions

L.F. conceived the research concept. L.F., M.Q.Z. supervised the research. Z.Y.Z., C.Y.W. carried out the main experiments, collected and analyzed the data. F.D. supervised the theoretical calculations, and M.J.S. performed the computational simulations. M.Y.D., X.T., D.L., and D.H.Z. contributed to the catalytic performance evaluation. S.Y.H. contributed to sample preparation and data analysis. Z.J.L., Y.S.M. processed the XAFS results. Y.L.Z. performed transmission electron microscopy characterizations. L.L. contributed to the in situ XRD characterizations. L.F., M.Q.Z., Z.Y.Z., C.Y.W., and M.J.S. cowrote the manuscript. All the authors contributed to data analysis and scientific discussion.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains Supplementary material available at [redacted].

Correspondence and requests for materials should be addressed to Mingyue Ding, Feng Ding or Lei Fu.

Peer review information *Nature Communications* thanks the anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.